

Multi-Criteria Decision-Making in Football: A Literature Review

Quantitative Decision-Making Course Project

1. Introduction

Football clubs make complex decisions every week, and most of them involve weighing several criteria at once. Buying a striker means thinking about goals, age, fitness, market value, injury history, and how the player fits the squad's tactical setup. Picking a starting eleven involves balancing form, opponent matchups, and rotation. These problems are exactly the kind that Multi-Criteria Decision-Making (MCDM) and Multi-Attribute Decision-Making (MADM) methods were designed for: ranking alternatives when no single metric tells the whole story.

Over the past two decades, researchers have applied MCDM techniques across nearly every football decision domain, from player recruitment to host-city selection. This review summarizes the main application areas, the methods used most often, and the gaps that still exist.

2. Player Selection and Recruitment

Player selection dominates the literature. Tavana et al. (2013) built one of the earliest fuzzy systems for evaluating players across technical, physical, tactical, and psychological criteria. Qader et al. (2017), publishing in *Measurement*, used TOPSIS to rank Malaysian U-17 players based on twelve measurable attributes including sprint times, anthropometric data, and skill tests. Their paper has become one of the most-cited football MCDM studies.

More recent work has shifted toward hybrid and fuzzy approaches. Pehlivan, Ünal, and Kahraman (2019) combined Fuzzy AHP with Fuzzy TOPSIS for national-team selection. Boran, Boran, and Dizdar (2024) used SWARA-VIKOR with neutrosophic numbers and notably distinguished between goalkeeper criteria (21 attributes including reflexes, jumping ability, and command of area) and outfield-player criteria (26 attributes covering marking, passing, dribbling, and footwork). Almutairi et al. (2025) recently published a Fuzzy AHP–Fuzzy TOPSIS framework covering twelve player criteria.

For transfer-window decisions specifically, Nasiri et al. (2019) developed a hybrid PROMETHEE–DEA model that explicitly accounts for budget constraints. Salles et al. (2022) combined a multiple-choice MCDM model with a genetic algorithm for Brazilian club squad formation. Ozceylan (2016) coupled AHP-derived weights with 0-1 integer programming to assemble a complete soccer squad.

3. Team Performance and Club Ranking

Team-level evaluation is the second largest sub-area. Galariotis, Germain, and Zopounidis (2018) ranked French Ligue 1 clubs simultaneously on business, financial, and sporting criteria using PROMETHEE II. Chelmiss et al. (2019) applied the same method to the Greek Superleague. Gökgöz and Yalçın (2021, 2023) have published extensively on World Cup and Champions League team evaluation, often combining Shannon Entropy weighting with TOPSIS, WASPAS,

or MAIRCA. Arsu (2021) used a bi-objective DEA approach for European clubs. The dual-axis structure of sporting criteria (xG, possession, goals scored and conceded) plus financial criteria (revenue, debt, attendance, broadcasting rights) has become standard.

4. Match Prediction and Other Application Areas

MCDM-based match prediction is less common than machine learning, but a few studies stand out. Saeed, Saqlain, and Riaz (2019) used neutrosophic soft sets with fuzzy TOPSIS to predict the FIFA 2018 World Cup winner. Principe et al. (2017) used probabilistic preferences for the Premier League. These approaches generally don't beat random forests or neural networks on raw accuracy, but they offer interpretability that black-box models can't.

Talent identification adds a forecasting dimension to player selection. Khan et al. (2024) applied AHP and TOPSIS to attackers, defenders, and midfielders using WhoScored data. The well-documented problem of scout disagreement (Lath et al., 2023, found very low inter-rater reliability among youth scouts) is a strong motivation for systematic MCDM frameworks.

Coach and manager selection has received surprisingly little dedicated attention despite its financial importance, with most studies treating managerial appointments as a sub-problem inside team evaluations. Referee assessment is similarly underdeveloped, though De Oliveira, Steffen, and Trojan (2023) reviewed the VAR literature and pointed toward MCDM as a natural framework. For tournament hosting, Górecka (2020) applied PROMETHEE and ELECTRE to host-city selection. Mu (2016) made a notable contribution with an AHP analysis arguing that Lionel Messi was not the most deserving winner of the 2014 FIFA Golden Ball. Görcün (2021) developed a CRITIC-WASPAS framework specifically for goalkeeper selection, addressing AHP's weaknesses with consistency burden and alternative count.

5. Methods and Trends

AHP and TOPSIS are the two most-used methods by a wide margin. PROMETHEE dominates outranking applications. WASPAS, CODAS, MABAC, and MAIRCA are all gaining ground in papers from 2020 onward. On the weighting side, AHP remains the standard for subjective expert weights, while CRITIC and Shannon Entropy are the preferred objective alternatives. Best-Worst Method (Rezaei, 2015) and SWARA are growing in popularity as well.

Fuzzy extensions are now standard whenever experts are involved. Triangular fuzzy numbers were the earliest approach, but intuitionistic, Pythagorean, hesitant, spherical, and neutrosophic variants have all appeared in football applications since 2018. Hybrid models are the dominant pattern: pick a weighting method (subjective like AHP, SWARA, or BWM, or objective like CRITIC or Entropy), pick a ranking method (TOPSIS, VIKOR, WASPAS, CODAS, or PROMETHEE), and run sensitivity analysis. Recent papers often apply multiple ranking methods in parallel and report rank correlations between them.

Geographically, the field is concentrated in a few research clusters. Turkish authors (Boran, Gökgöz, Yalçın, Pehlivan, Kahraman) have produced a disproportionate share of the literature. Brazilian, Polish, Iranian, and Greek groups also contribute heavily.

6. Gaps and Conclusion

Several gaps stand out in the current literature. Women's football is almost entirely absent from MCDM studies. Coach selection lacks dedicated research despite the financial stakes involved. Validation against actual transfer outcomes (does the recommended signing actually perform better?) is rare; most papers stop at internal sensitivity analysis. Position-specific criteria are inconsistent across studies, with Boran et al. (2024) being one of the few systematic exceptions. Integration with modern event-stream tracking data from providers like Opta and StatsBomb is also limited.

For practical applications such as transfer-window decision-support tools, the literature points to a clear design pattern: combine subjective and objective weights (AHP plus CRITIC works well), run multiple ranking methods in parallel (TOPSIS, VIKOR, WASPAS, and CODAS together), report disagreement when methods diverge, and use position-specific criterion sets. This matches the methodological consensus that has emerged in the past five years and is a sensible foundation for any squad-builder application.

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